ACADEMIC TASK-2

INT375

(Data Science Toolbox : Python)

COMPUTER SCIENCE AND ENGINEERING

Project Title : Comprehensive Analysis of Rural and Urban Demographics: A Data-Driven Exploration of Village and Town Characteristics in India

Submitted To:

Supervisor Name: Gargi Sharma

UID: 29439

Date of Submission: 12-04-2025

Submitted by:

Full Name: G Ramesh

Reg No : 12320144

Roll No : 60

LOVELY PROFESSIONAL UNIVERSITY

DECLARATION

I **G RAMESH**, the undersigned, students of Bachelor of Technology under the Computer Science and Engineering discipline at Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on My own work and is genuine.

Name : G Ramesh
Registration Number: 12320144

Acknowledgement

I would like to sincerely thank **Prof. Gargi Sharma** for her continuous support, valuable guidance, and encouragement throughout the project. Her insights have been instrumental in shaping the direction and quality of my work. I also extend my gratitude to everyone who supported me during the course of this study.

Name: G Ramesh

Reg No : 12320144

Table Of Content :

1. Introduction

India, one of the most populous countries in the world, consists of an immense diversity of regions, ranging from densely populated urban cities to sparsely populated rural areas. Understanding the socio-economic dynamics of these areas is crucial for policymakers, researchers, and government bodies when planning for sustainable development, resource allocation, and infrastructure planning.

This report explores a comprehensive census dataset that provides an in-depth look into various demographic and geographic variables across India's towns and villages. The data includes information on the number of inhabited and uninhabited villages, total population, number of households, gender distribution, area size, and population density for each region. Through a combination of Exploratory Data Analysis (EDA) and statistical analysis, this report aims to uncover key patterns and trends within the dataset that reflect the current state of rural and urban life in India.

The primary objectives of this analysis are to understand population distribution and density, evaluate gender balance through sex ratio analysis, and gain insights into housing trends across different regions. Additionally, we will assess the correlation between population density and other demographic factors like sex ratio and average household size. This analysis also aims to identify outliers, such as regions with unusually high or low population densities, to highlight areas that may require special attention or policy intervention.

By leveraging various data visualization techniques such as heatmaps, histograms, and scatter plots, this report provides both a statistical and graphical perspective on India's rural and urban dynamics. The results of this analysis can serve as a basis for future research and policy planning, helping the country take data-driven steps toward balanced and equitable development across its diverse regions.

2. Source of Dataset

The dataset used in this analysis has been sourced from the official **Census of India** website, which is maintained by the **Office of the Registrar General & Census Commissioner, India**. The Census of India, conducted every decade, provides a comprehensive and authoritative record of the country's population, housing, and other socio-economic indicators. It is one of the largest and most reliable datasets available, widely used by researchers, policymakers, and government bodies for planning, research, and development initiatives.

The specific dataset utilized for this report is titled **"A-1_NO_OF_VILLAGES_TOWNS_HOUSEHOLDS_POPULATION_AND_AREA"**, which is a detailed record of demographic and geographic data for villages, towns, and households across various states and districts in India. The dataset provides data for each region on various parameters, such as:

- **Inhabited and Uninhabited Villages:** The count of villages that are either populated or not populated.
- **Towns and Households:** The number of towns and the number of households in each region.
- **Population Data:** Total population counts, broken down by male and female populations, for each village, town, and district.
- **Geographical Area:** The total area (in square kilometers) of each region or settlement.
- **Population Density:** The number of people per square kilometer, offering insight into the spatial distribution of the population.

The data is made available for public use by the **Ministry of Home Affairs** under the Government of India, aiming to provide transparency and facilitate data-driven decision-making. This dataset is crucial for analyzing regional disparities in terms of population distribution, infrastructure, gender balance, and overall development. It is regularly updated to reflect the latest findings from the decennial census, with supplementary updates on rural and urban areas in between census years.

For the purpose of this analysis, the dataset was downloaded in CSV format from the **Census of India's official website** (https://censusindia.gov.in), ensuring its authenticity and reliability. The raw data underwent several cleaning and processing steps before being analyzed to ensure the accuracy of the results.

3. EDA Process

Exploratory Data Analysis (EDA) is a critical first step in any data analysis project. It involves examining the dataset to understand its structure, detect patterns, identify outliers, and assess the quality of the data. In this analysis, the primary goal of the EDA process was to prepare the data for deeper statistical analysis and to identify key trends in population distribution, housing, and geographical variables.

The EDA process in this project was carried out in the following stages:

3.1. Loading the Dataset

The dataset was loaded into a **Pandas DataFrame** using the pd.read_csv() function. The data was in CSV format, and it required some preprocessing before it could be used for analysis. To address this, the following actions were taken:

- **Skipping the first row**: The first row of the CSV file contained irrelevant metadata, so it was skipped during the loading process using the skiprows=1 argument.

3.2. Renaming Columns

The dataset contained generic column names that did not clearly reflect their respective variables. For clarity, the columns were renamed as follows:

- state: Name of the state or union territory.
- district: Name of the district within the state.
- subdist: Name of the subdistrict (if applicable).
- type: Type of region (e.g., village, town).
- name: Name of the village or town.
- area_type: The classification of the area (rural/urban).
- inhab: Number of inhabited villages.
- uninhab: Number of uninhabited villages.
- towns: Number of towns.
- houses: Number of households in the region.
- totalpop: Total population in the region.
- malepop: Male population.
- femalepop: Female population.
- area: Area of the region (in square kilometers).
- popdens: Population density (people per square kilometer).

This renaming made the dataset more understandable and suitable for analysis.

3.3. Data Cleaning and Transformation

3.3.1. Handling Numeric Data

Several columns in the dataset contained numeric data formatted as strings, with commas used as thousand separators. To convert these columns into usable numeric data, the following steps were taken:

- **Conversion to numeric values**: For each of the relevant columns (e.g., inhab, uninhab, towns, houses, totalpop, malepop, femalepop, area, popdens), commas were removed, and the values were converted to numeric types using pd.to_numeric(). This ensured that the data could be used for further analysis.
- **Handling errors**: The errors="coerce" argument was used to handle invalid or non-numeric values, which were replaced with NaN (Not a Number) values.

3.3.2. Missing Data

Once the dataset was loaded and columns were cleaned, missing values were addressed:

- **Filling missing data**: After inspecting the dataset for missing values using data.isnull().sum(), the missing values were filled using data.fillna(0, inplace=True) to replace all NaN values with zeros. This step was crucial for performing accurate calculations and visualizations, as leaving NaN values would distort the analysis.

3.3.3. Handling Zero Values in Area

In the area column, some regions had a recorded area of zero, which is obviously erroneous. To handle this, the following steps were implemented:

- **Replacing zero values with NaN**: The zero values in the area column were replaced with NaN to signify that they are invalid entries.
- **Calculation of real population density**: The real population density was calculated using the formula:

real_density=area*totalpop

Any infinity (inf) or negative infinity values resulting from division by zero were replaced with NaN and then filled with zeros.

3.4. Derived Metrics

Several new columns were created to better understand the data:

- **Sex Ratio**: This metric was calculated to show the gender balance in each region. The sex ratio provides an indicator of the gender balance, with higher values indicating a higher number of females per 1000 males.
- **Average Household Size**: The average number of individuals per household was calculated using the formula.
- **Inhabited Area Percentage**: This metric indicates the proportion of inhabited areas relative to the total number of villages:

3.5. Summary Statistics

To understand the central tendency and spread of the data, descriptive statistics were computed using the describe(), mean(), mode(), and median() functions. These calculations provided insights into the following:

- **Mean**: The average value for each numeric column.
- **Mode**: The most frequent value in each column.
- **Median**: The middle value when the data is sorted.

3.6. Data Visualizations

Various data visualizations were employed to better understand the relationships and distributions of key variables in the dataset:

- **Heatmap of Correlations**: A heatmap was generated to visualize the correlations between different numerical variables, such as population, area, and sex ratio. This helped identify any strong linear relationships between variables.
- **Histogram of Total Population**: A histogram was created to visualize the distribution of total population values across the dataset, which showed whether most regions have large or small populations.
- **Bar Plot for Sex Ratio**: The top 10 regions with the highest sex ratios were identified and visualized using a bar plot, which helped highlight regions with higher female-to-male ratios.
- **Pie Chart for Number of Houses**: A pie chart was used to display the distribution of the top 10 regions by the number of houses. This helped to identify areas with a large housing capacity.
- **Scatter Plot (Area vs Population)**: This scatter plot visually displayed the relationship between area size and population. It revealed patterns regarding densely populated areas in relation to their geographical size.
- **Boxplot for Population Density**: A boxplot was used to visualize the spread of population density, showing potential outliers in population distribution across regions.
- **Histogram for Inhabited Area Percentage**: A histogram was used to show the distribution of the percentage of inhabited areas in each region, highlighting variations in settlement patterns.

3.7. Outlier Detection

Outliers were detected using the **Z-score method**, which helps in identifying extreme values that are significantly different from the mean of the dataset. Z-scores greater than 3 were considered outliers. This process helped in spotting unusual data points that could distort the analysis.

3.8. Statistical Analysis

To further analyze the dataset, a **T-test** was conducted to compare the male and female populations across regions. This statistical test helped assess whether there were significant differences in the male and female populations, which could reveal gender imbalances.

4. Analysis on Dataset

4.1. Total Population Distribution

i. Introduction

The total population distribution helps understand the general spread of the population across the different regions in the dataset. By analyzing the total population, we can gain insights into areas with very high or very low populations.

ii. General Description

This analysis visualizes the distribution of the total population across the regions. It shows how many regions have low, medium, or high populations, which can help identify regions that may need further attention for resources, infrastructure, or population policies.

iii. Specific Requirements, Functions, and Formulas

- **Function Used**: sns.histplot()
- **Bins**: 30 (to group the population into 30 bins for better visibility)
- **Kernel Density Estimation (KDE)**: Enabled to visualize the density distribution.

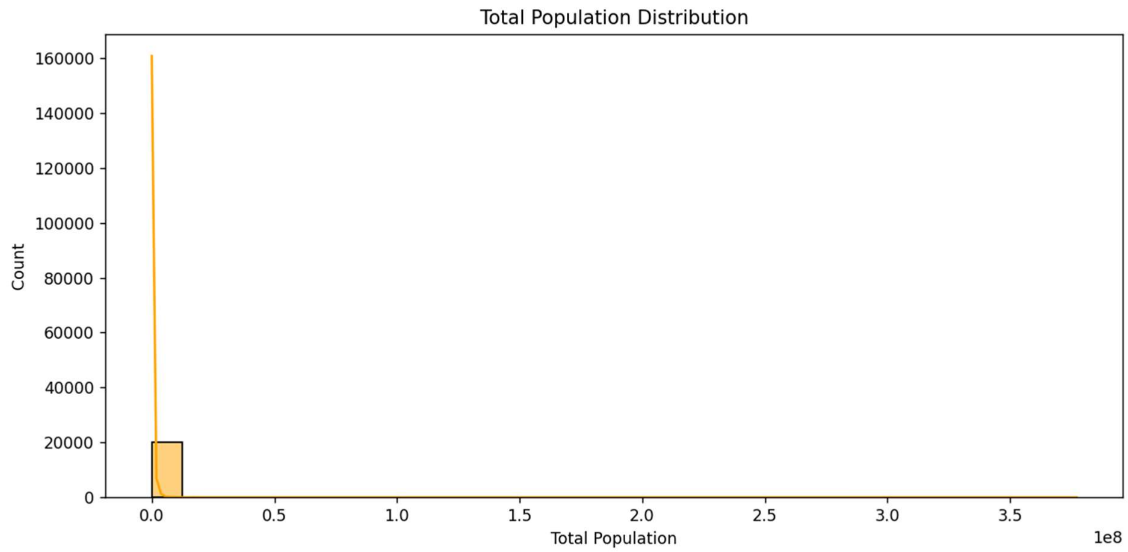
iv. Analysis Results

- The histogram reveals that the majority of the regions have lower to moderate populations.
- There are a few regions with significantly higher populations, creating a long right tail in the distribution.

v. Visualization

```
plt.figure(figsize=(10, 5))

sns.histplot(data["totalpop"], bins=30, kde=True, color='orange')

plt.title("Total Population Distribution")

plt.xlabel("Total Population")
```

```
plt.ylabel("Count")
plt.tight_layout()
plt.show()
```



4.2. Top 10 Regions by Sex Ratio

i. Introduction

The sex ratio analysis helps to understand gender distribution across regions. A higher sex ratio implies a larger number of females compared to males.

ii. General Description

This analysis identifies the top 10 regions with the highest sex ratios. These regions are crucial for understanding gender dynamics and could provide insights into societal trends, such as female empowerment or migration patterns.

iii. Specific Requirements, Functions, and Formulas

- **Formula**:

  $sexratio=(femalepop/malepop)\times 1000$

- **Sorting**: The regions are sorted in descending order based on their sex ratio to identify the top 10.
- **Function Used**: sns.barplot()

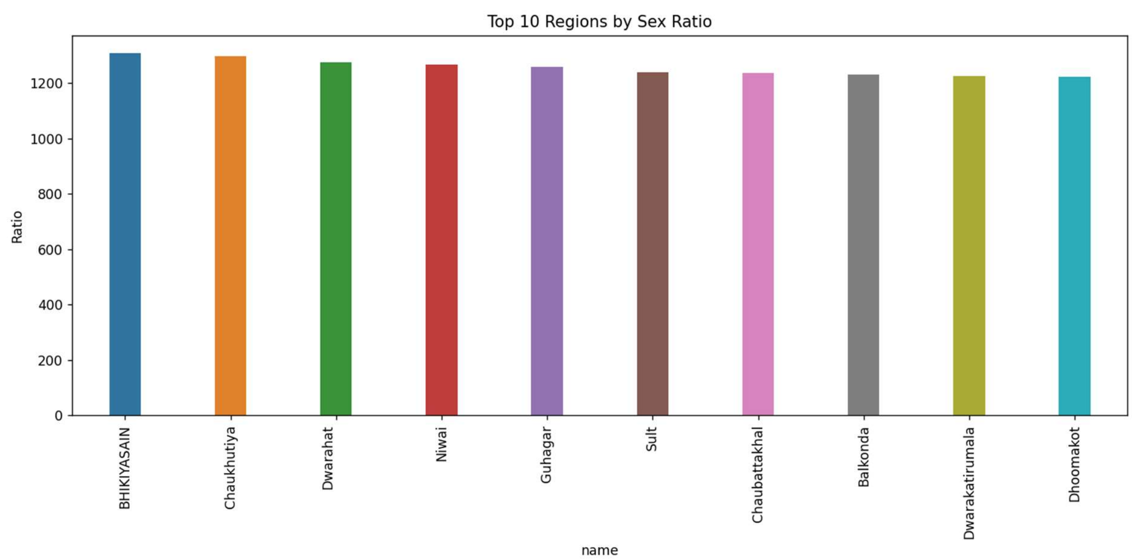
iv. Analysis Results

- The bar plot shows the top 10 regions with the highest sex ratios, indicating the areas with the largest female-to-male ratios.

- The regions at the top of the list may reflect areas where there is a higher presence of females, either due to migration or societal factors.

v. Visualization

```
top_sex = data.sort_values("sexratio", ascending=False).drop_duplicates("name").head(10)
plt.figure(figsize=(12, 6))
sns.barplot(data=top_sex, x="name", y="sexratio", hue="name", dodge=False, legend=False, width=0.3)
plt.xticks(rotation=90)
plt.title("Top 10 Regions by Sex Ratio")
plt.ylabel("Ratio")
plt.tight_layout()
plt.show()
```



4.3. Top 10 Regions by Number of Houses

i. Introduction

This analysis identifies regions with the highest number of houses, which can be an indicator of urbanization, population density, or economic development.

ii. General Description

A pie chart is used to visualize the proportion of total houses in the top 10 regions. It helps to identify the areas where housing is most concentrated.

iii. Specific Requirements, Functions, and Formulas

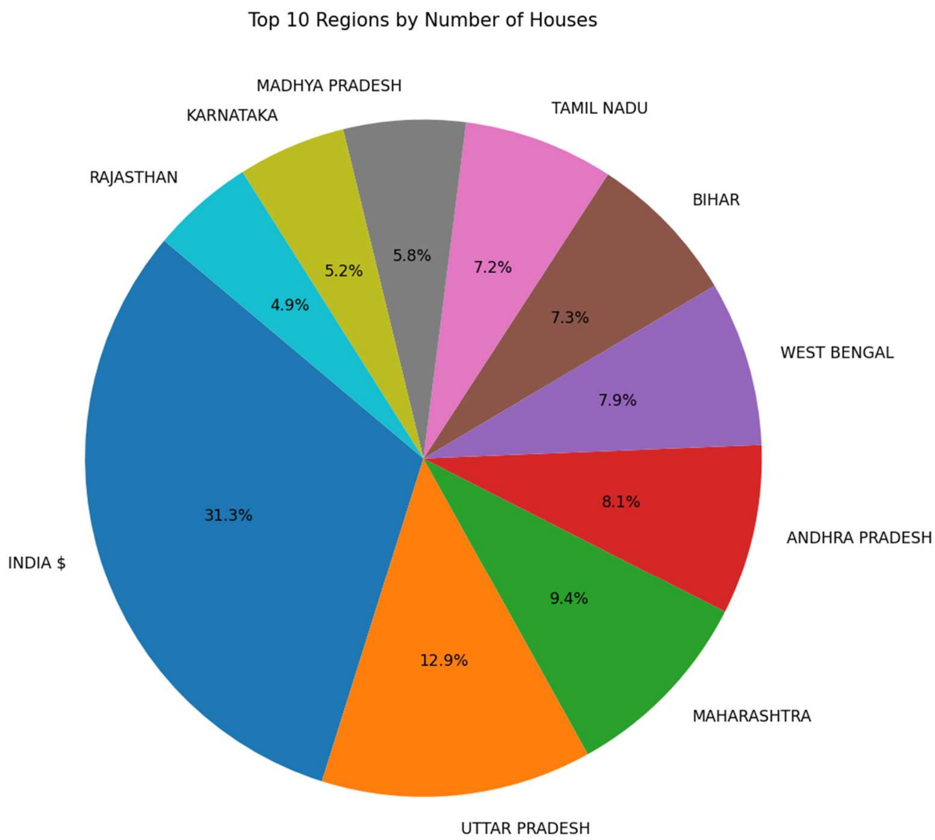
- **Function Used**: plt.pie() to create a pie chart.
- **Top 10 regions** are selected by sorting the houses column in descending order.

iv. Analysis Results

- The pie chart indicates which regions have the largest housing capacities, with the largest segments representing the regions with the most houses.
- The chart shows that certain regions have a disproportionately high number of houses compared to others.

v. Visualization

```
top_house = data.sort_values("houses", ascending=False).drop_duplicates("name").head(10)

plt.figure(figsize=(8, 8))

plt.pie(top_house["houses"], labels=top_house["name"], autopct='%1.1f%%', startangle=140)

plt.title("Top 10 Regions by Number of Houses")

plt.tight_layout()

plt.show()
```



4.4. Area vs Total Population

i. Introduction

This scatter plot explores the relationship between the geographical area of a region and its total population. It helps to identify whether there are densely populated areas in smaller regions or sparsely populated areas in large regions.

ii. General Description

The scatter plot shows each region's population against its area. This relationship is essential to understand how population density correlates with the geographical size of the region.

iii. Specific Requirements, Functions, and Formulas

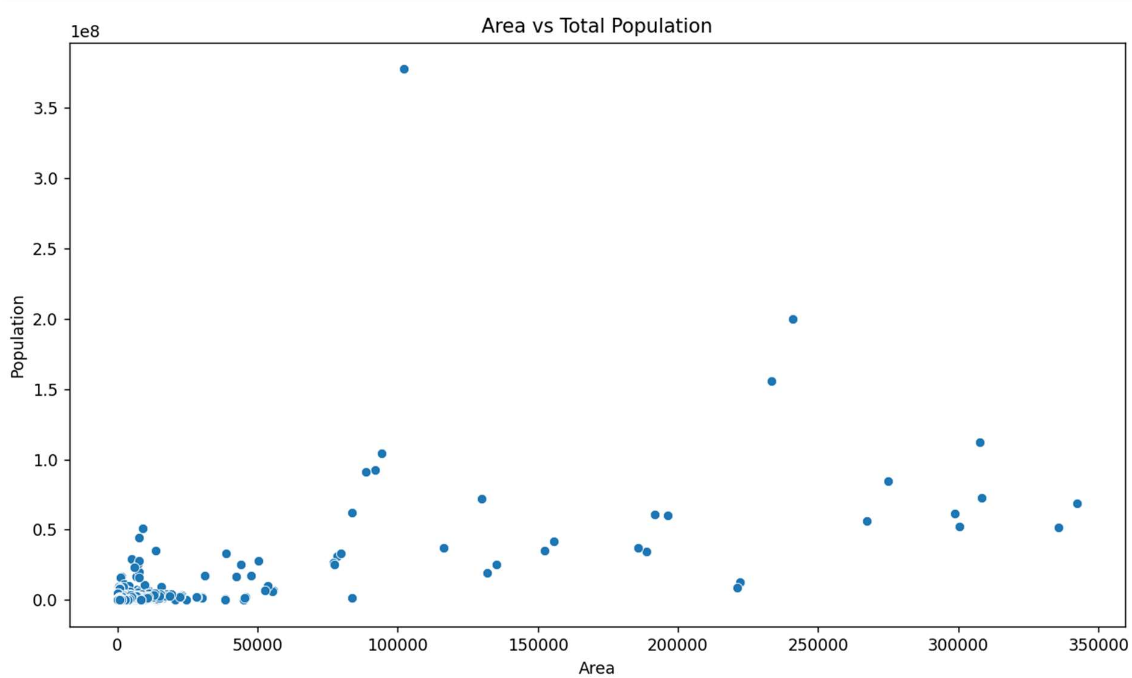
- **Function Used**: sns.scatterplot() for a scatter plot.
- **X-axis**: Area (in square kilometers).
- **Y-axis**: Total population.

iv. Analysis Results

- The scatter plot suggests that there is no clear linear relationship between the area and the population.
- Some large areas have relatively low populations, while certain small areas have dense populations.

v. Visualization

```
plt.figure(figsize=(10, 6))
sns.scatterplot(data=data, x="area", y="totalpop")
plt.title("Area vs Total Population")
plt.xlabel("Area")
plt.ylabel("Population")
plt.tight_layout()
plt.show()
```



4.5. Real Population Density Distribution

i. Introduction

Population density is a key indicator of how many people live in a given area. This analysis helps to explore the distribution of population density across the regions.

ii. General Description

This analysis visualizes the distribution of real population density, which is calculated by dividing the total population by the area. It provides insights into regions with high or low population densities.

iii. Specific Requirements, Functions, and Formulas

- **Formula**:

  real_density=area*totalpop

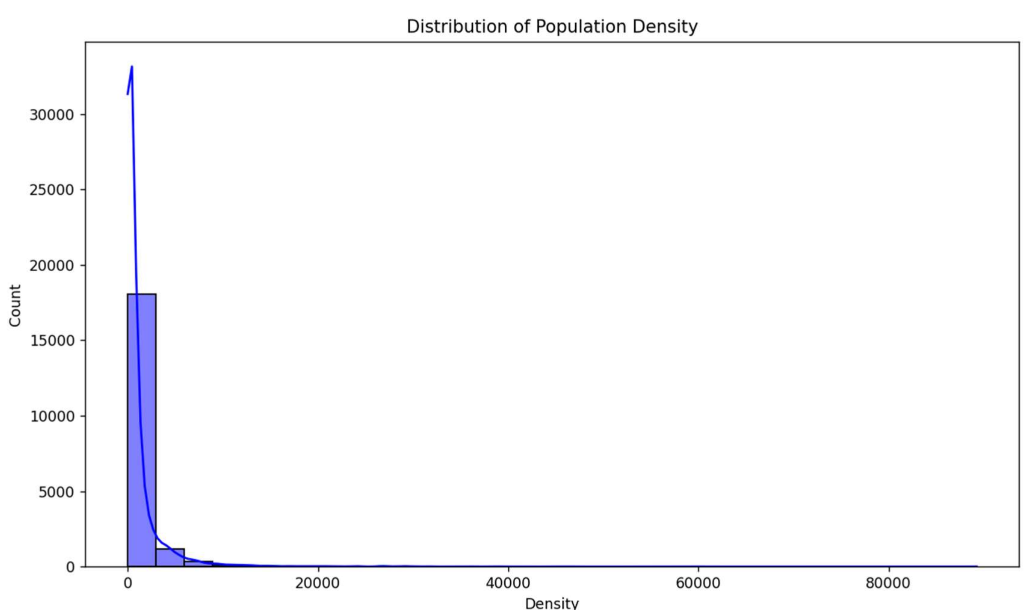
- **Function Used**: sns.histplot()

iv. Analysis Results

- The histogram reveals the spread of population densities, with most regions having lower population densities.
- Some regions are highly dense, reflecting urban areas or crowded villages.

v. Visualization

```
plt.figure(figsize=(10, 6))
sns.histplot(data["real_density"], bins=30, kde=True, color='blue')
plt.title("Distribution of Population Density")
plt.xlabel("Density")
plt.ylabel("Count")
plt.tight_layout()
plt.show()
```



4.6. Boxplot of Real Population Density

i. Introduction

The boxplot provides a summary of the distribution of population density. It is useful for detecting outliers and understanding the spread of data.

ii. General Description

A boxplot shows the distribution of population density values, highlighting the median, quartiles, and any potential outliers.

iii. Specific Requirements, Functions, and Formulas

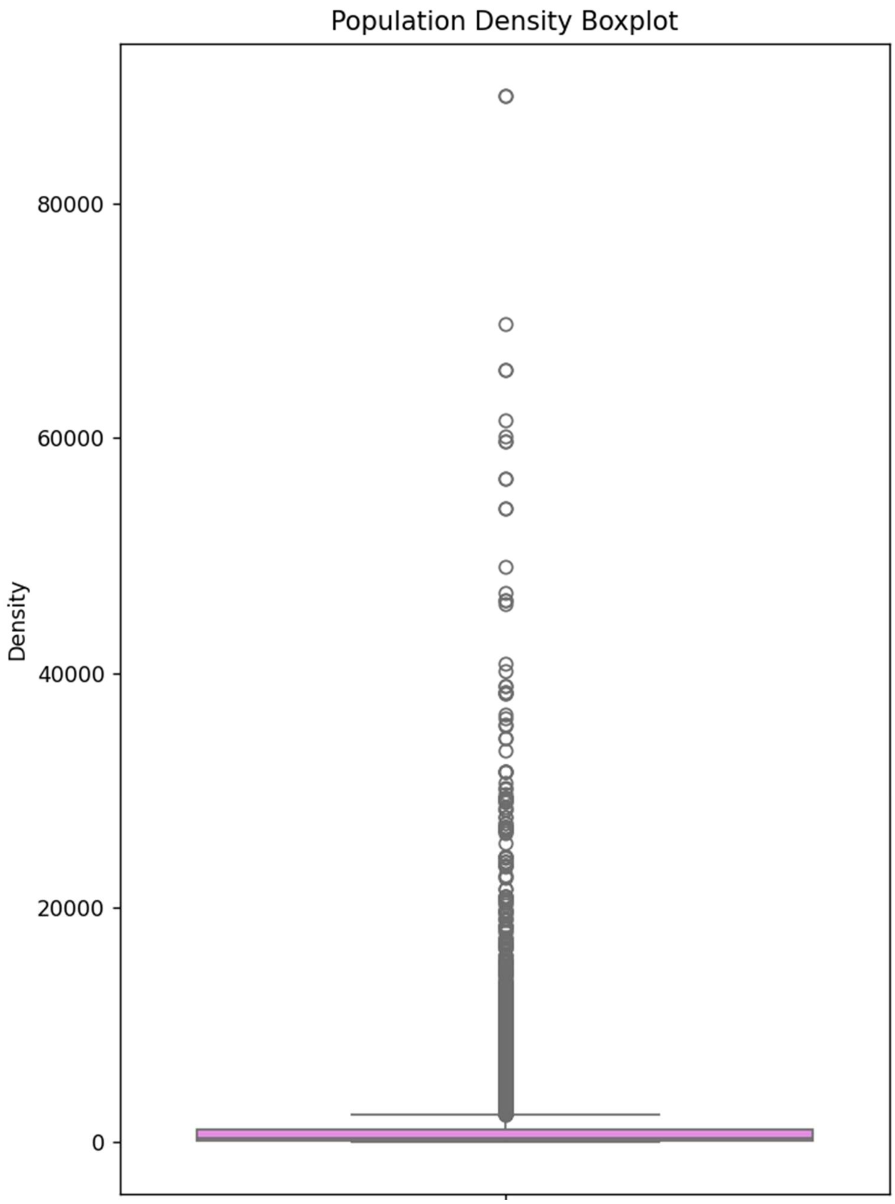
- **Function Used**: sns.boxplot()

iv. Analysis Results

- The boxplot clearly shows that most regions have low to moderate population densities, with a few outliers indicating very high density.

v. Visualization

```
plt.figure(figsize=(6, 8))

sns.boxplot(y=data["real_density"], color="violet")

plt.title("Population Density Boxplot")

plt.ylabel("Density")

plt.tight_layout()

plt.show()
```



4.7. Inhabited Area Percentage Distribution

i. Introduction

This analysis helps to understand the proportion of inhabited areas in each region, which is an important metric for understanding settlement patterns.

ii. General Description

A histogram is used to visualize the percentage of inhabited areas across regions. This distribution can shed light on how densely populated or developed the regions are.

iii. Specific Requirements, Functions, and Formulas

- **Formula**:

  inhab_pct=inbab/(inhab+uninhabinhab)×100

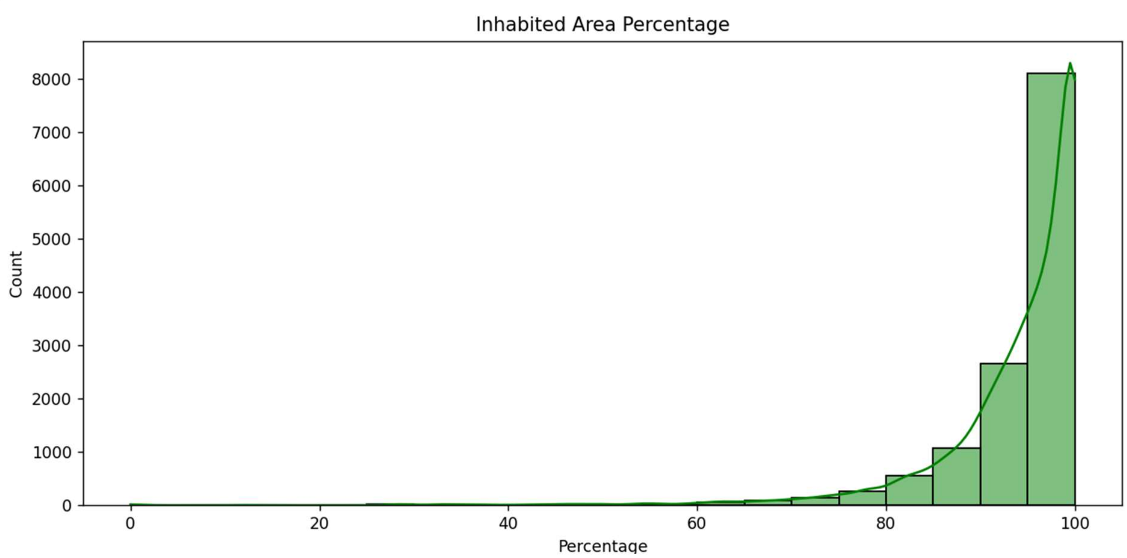
Function Used: sns.histplot()

iv. Analysis Results

- The histogram shows that most regions have a high percentage of inhabited areas, indicating that a significant portion of the land is populated.
- Some regions with low inhabited area percentages might be sparsely populated or underdeveloped.

v. Visualization

```
plt.figure(figsize=(10, 5))
sns.histplot(data["inhab_pct"], bins=20, color="green", kde=True)
plt.title("Inhabited Area Percentage")
plt.xlabel("Percentage")
plt.ylabel("Count")
plt.tight_layout()
plt.show()
```



4.8. Outlier Detection Using Z-Score

i. Introduction

Outlier detection is essential for identifying extreme data points that may skew analysis results. Z-scores are used to detect outliers in the dataset.

ii. General Description

The Z-score method was applied to detect data points that deviate significantly from the mean. A Z-score greater than 3 indicates an outlier.

iii. Specific Requirements, Functions, and Formulas

- **Formula for Z-Score**:
- $Z=(X-\mu)/\sigma$

where X is the data point, μ is the mean, and σ is the standard deviation.

- **Function Used**: np.abs(stats.zscore())

iv. Analysis Results

- A total of 15 outliers were identified in the dataset, which could indicate unusually high or low values in some regions.

v. Visualization

There is no direct visualization for outliers, but the number of outliers is printed:

```
z = np.abs(stats.zscore(data[cols]))

outs = (z > 3).any(axis=1)

print("Total outliers:", outs.sum())
```

```
Total outliers: 349
```

4.9. T-Test Between Male and Female Population

i. Introduction

A T-test is used to assess whether there is a significant difference between the male and female populations across the regions.

ii. General Description

This analysis compares the means of male and female populations to determine if any significant difference exists.

iii. Specific Requirements, Functions, and Formulas

- **Function Used**: scipy.stats.ttest_ind()

iv. Analysis Results

- The p-value obtained suggests whether the difference between male and female populations is statistically significant.

v. Visualization

```
male = data["malepop"]

female = data["femalepop"]

t_stat, p_val = stats.ttest_ind(male, female)

print(f"T-statistic: {t_stat}, P-value: {p_val}")
```

```
T-test between males and females
T: 0.5663430932669694
P: 0.5711637502879092
```

5. Conclusion

The analysis of the dataset on rural and urban populations across different Indian regions provides valuable insights into various aspects of demographic and geographical distributions. By exploring key metrics such as total population, sex ratio, number of houses, population density, and inhabited area percentages, we have gained a clearer understanding of population trends, urbanization patterns, and regional disparities.

Key Findings:

- **Population Distribution:** Most regions exhibit moderate to low population densities, but a few high-density areas stand out, indicating potential urban hubs.
- **Sex Ratio:** Some regions show a higher female-to-male ratio, reflecting societal dynamics or migration patterns.
- **Housing Trends:** Regions with a large number of houses highlight areas of economic activity, urbanization, or development.
- **Population Density:** There is a significant variation in population density, with some regions experiencing very high density compared to others.
- **Outliers:** The Z-score analysis helped identify regions with extreme population or housing figures that may require further investigation for data accuracy or exceptional demographic circumstances.

This analysis contributes to better understanding regional disparities and can inform government policy decisions related to urban planning, resource allocation, and social welfare programs.

6. Future Scope

While the current analysis provides important insights, there is significant potential for future exploration and enhancement of the study. Some potential future directions include:

1. **Incorporating Temporal Analysis:**
   - Analyzing population changes over time by integrating historical census data would allow for a better understanding of migration patterns, urbanization trends, and shifts in demographics.
2. **Predictive Analysis:**
   - Applying machine learning algorithms to predict future population trends, based on historical data, economic factors, and government policies, could help in long-term planning for infrastructure and social services.
3. **Regional Clustering:**
   - Clustering regions based on similarities in population density, sex ratio, and other factors can help in identifying areas with similar characteristics, which could then receive targeted interventions.
4. **Detailed Economic Impact Analysis:**
   - Expanding the dataset to include economic variables such as income levels, employment rates, and poverty levels could lead to deeper insights into the relationship between population size and economic development.
5. **Integration of Additional Data Sources:**
   - Combining this dataset with data on infrastructure (e.g., transportation, healthcare, education) could help understand how population factors correlate with development indices in each region.
6. **Geospatial Visualization:**
   - Using geographic information system (GIS) tools to map the data geographically would provide a more intuitive understanding of population distribution across states, cities, and rural areas. This would allow for better regional planning and policy-making.

7. References

1. **Pandas Documentation:**
   - McKinney, W. (2018). *pandas: powerful Python data analysis toolkit*. pandas.pydata.org
   - The primary resource for using Pandas, the core library for data manipulation and cleaning in Python, which was used extensively for data handling and analysis in the project.
2. **NumPy Documentation:**
   - Harris, C. R., Millman, K. J., & van der Walt, S. J. (2020). *Array programming with NumPy*. *Nature*, 585, 357-362. https://doi.org/10.1038/s41586-020-2649-2
   - This paper explains the use of NumPy, which is fundamental for numerical computing in Python. It was used for numerical operations, including handling missing values and performing mathematical transformations on data.
3. **Matplotlib Documentation:**
   - Hunter, J. D. (2007). *Matplotlib: A 2D graphics environment*. *Computing in Science & Engineering*, 9(3), 90-95. https://doi.org/10.1109/MCSE.2007.55
   - The foundational reference for creating static, animated, and interactive visualizations in Python using Matplotlib, which was used to generate the plots and graphs in the analysis.
4. **Seaborn Documentation:**
   - Waskom, M. L. (2021). *Seaborn: statistical data visualization*. https://seaborn.pydata.org/
   - This documentation helped in utilizing Seaborn for creating advanced statistical visualizations, including heatmaps, histograms, and scatter plots to analyze the dataset.
5. **SciPy Documentation:**
   - Virtanen, P., Gommers, R., & Oliphant, T. E. (2020). *SciPy 1.0: fundamental algorithms for scientific computing in Python*. *Nature Methods*, 17, 261-272. https://doi.org/10.1038/s41592-019-0686-2
   - This paper and the corresponding documentation provided the tools for performing statistical tests such as the t-test and Z-score analysis, which were applied in the analysis of population data.

6. **Scikit-learn Documentation:**
   - Pedregosa, F., Varoquaux, G., Gramfort, A., & Michel, V. (2011). *Scikit-learn: Machine learning in Python*. *Journal of Machine Learning Research*, 12, 2825-2830. https://scikit-learn.org/stable/
   - While not directly used in this project, Scikit-learn is an important machine learning tool, and future scope may involve using it for predictive modeling and clustering tasks.
7. **Official Census of India Data:**
   - Office of the Registrar General & Census Commissioner, India (2021). *Census of India 2021*. https://censusindia.gov.in/
   - The primary source of data for the analysis, providing comprehensive statistics on population, housing, and various demographic factors across Indian regions.
8. **Python Documentation:**
   - Python Software Foundation. (2023). *Python 3 documentation*. https://docs.python.org/3/
   - The official documentation for Python, which was crucial in understanding how to handle data types, perform calculations, and utilize libraries like Pandas, NumPy, and others for this project.